

Endpoint structure and a conditional nonlinear matching limit

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Result and scope. The calculation does not determine $M_{\text{ext}}(J, \alpha)$ at finite coupling. It derives the exact finite-coupling throat exponent, an irregular squashing equation about the critical static exterior, and a nonlinear intermediate connection to the inner BTZ region. Within an explicitly assumed matched limit, this gives

$$\Omega_H \sqrt{\alpha} \longrightarrow \frac{2}{3}, \quad M = \frac{3\pi\alpha}{4} + \frac{4J}{3\sqrt{\alpha}} + o(J).$$

The intermediate profile passes numerical residual and cutoff checks. Convergence of the full rotating global branch to this matched limit is not proved; the critical static mass remains a global endpoint hypothesis.

The inherited inputs are the reduced EGB action of [1] and the throat family of [2]. Our action is $R + \alpha\mathcal{L}_{\text{GB}}$, $G_5 = 1$, and J denotes the angular momentum in each plane. The symbolic reductions below were derived here; the Kaluza–Klein scalar-curvature identity used for the final throat curvature check was recalled.

The unique physical throat and its singular limit. Write $a = \alpha/g_H$, $q = h_H/g_H$, and use $g_H = 1$ temporarily. Define

$$P = 16a^2q^2(4 - q) - 8a(2q^2 - 7q + 4) + q - 2 = 0, \quad (1)$$

$$R = 1 + 4a(4 - 3q), \quad v = \frac{R[3 + 4a(4 - q)]}{2(4 - q)[3 + 4a(8 - 6q)]}, \quad k^2 = \frac{4(4 - q)v^2}{qR}, \quad (2)$$

$$\bar{J}^2 = \frac{8J^2}{\pi^2 g_H^3} = \frac{q^2(4 - q)R}{8}. \quad (3)$$

The discriminant is

$$\text{disc}_q P = 64a^2(4a + 1)[131072a^4 + 28672a^3 + 6528a^2 - 92a + 5] > 0 \quad (a > 0).$$

The last quadratic part is strictly positive. Together with the exact signs at $q = 0, 2, 4$ and $q_c = (1 + 16a)/(12a)$, this places one root in $(0, 2)$, a negative root, and one positive root between 4 and q_c . The latter has $\bar{J}^2 < 0$; at $a = 1/32$ it reaches the degenerate $q = 4$, $J = 0$ point. Thus only the root in $(0, 2)$ is physical for $J > 0$. The radical parametrization has $D(q) = 5q^4 - 34q^3 + 73q^2 - 56q + 16$ and $\text{disc } D = -3506176 \neq 0$.

For $t = (2a)^{-1/2}$, exact coefficient checks give

$$q = t - \frac{3}{4}t^2 + \frac{13}{32}t^3 - \frac{3}{32}t^4 + O(t^5), \quad (4)$$

$$v = a + \frac{5}{32} + \frac{3}{128}t + O(t^2), \quad (5)$$

$$\bar{J}^2 = 4 - 10t + \frac{63}{4}t^2 + O(t^3), \quad \frac{qk^2}{v} = 1 + \frac{1}{2}t + O(t^2). \quad (6)$$

At fixed α , this implies the local scalings

$$h_H \sim \frac{g_H^{3/2}}{\sqrt{2\alpha}}, \quad J \sim \frac{\pi}{\sqrt{2}} g_H^{3/2}, \quad v_{1,\text{physical}} \rightarrow \alpha.$$

The fibre–AdS₂ metric becomes locally AdS₃. The recalled curvature identity gives $g_H \mathcal{R}_H = 8 - 2q - 2/v + qk^2/(2v^2) \rightarrow 8$, so the $g_H \rightarrow 0$ throat endpoint is curvature singular. It gives no asymptotic mass.

Exact finite-coupling indicial equation. Fix $f = \rho^2/v$ and perturb the throat by $\delta b/b = \beta\rho^\lambda$, $\delta g/g = \eta\rho^\lambda$, $\delta h/h = \chi\rho^\lambda$, and $\delta w'/w' = \omega\rho^\lambda$. For the physical mode, the constraints imply

$$\frac{\chi}{\eta} = -\frac{2(1 + 4aq)}{R}, \quad \frac{\beta - 2\omega}{\eta} = \frac{4(48a^2q^2 + 24aq - 16a - 1)}{R^2}.$$

The remaining equation is

$$\lambda^2 + \lambda - Q(a, q) = 0, \quad Q = \frac{4 \sum_{n=0}^5 a^n N_n(q)}{(q-4)(1+8aq)(12aq-16a-1)^2(24aq-32a-3)}, \quad (7)$$

where

$$\begin{aligned} N_5 &= 2048q^2(6q^4 - 61q^3 + 212q^2 - 304q + 192), \\ N_4 &= 256q(-32q^4 + 159q^3 - 132q^2 - 48q + 320), \\ N_3 &= 128(-44q^4 + 213q^3 - 274q^2 + 208q + 32), \\ N_2 &= 64(4q^3 + 6q^2 - 27q + 44), \quad N_1 = -16(q^2 + 12q - 19), \quad N_0 = 9. \end{aligned}$$

All five maximal minors of the 5×4 matrix vanish modulo (1) and (7). The physical kinetic coefficient after elimination is $3v(1+8aq)/R$. Benchmarks are $Q(0, 2) = 6$ and $Q(1/12, 1) = 288/55$. Comparison with the independent saved numerical indicial calculation gives 63 matches with maximum absolute error 1.981×10^{-13} .

The positive root γ satisfies

$$Q = \frac{3}{t} + 1 + \frac{47}{32}t + O(t^2), \quad \gamma \sim \sqrt{3}(2a)^{1/4} - \frac{1}{2}.$$

It returns to $\gamma = 2$ at $a = 0.73192922516721036349 \dots$, $q = 0.49751074328671794486 \dots$, verified to 60 decimal places by (1) and $Q = 6$. Therefore an uncanceled $\Gamma(2 - \gamma)$ factor would have an interior pole. This does not exclude expressions in which such poles cancel.

Critical-static exterior: the missing squashing layer. The inherited static endpoint metric has

$$N(r) = 1 + \frac{r^2 - \sqrt{r^4 + 16\alpha^2}}{4\alpha}, \quad \mu = 2\alpha, \quad M_* = \frac{3\pi\alpha}{4}.$$

Introduce the angular traceless deformation $g = r^2 e^{-\eta/3}$, $h = r^2 e^{2\eta/3}$, with $b = f = N$. The quadratic reduced action derives the normal equation

$$(A\eta')' - V\eta = 0, \quad A = \frac{Nr^5}{\sqrt{r^4 + 16\alpha^2}}, \quad V = \frac{8r^3(r^4 + 48\alpha^2)}{(r^4 + 16\alpha^2)^{3/2}}. \quad (8)$$

All five full linearized Euler equations were checked: the b, f, w equations vanish and the g, h equations are exactly (8). Both A, V are positive for $r > 0$.

The normal form and its leading solutions at the singular end are

$$\eta'' + \left(\frac{7}{r} - \frac{r}{4\alpha} + O(r^3) \right) \eta' + \left(-\frac{96\alpha}{r^4} - \frac{12}{r^2} + \frac{5}{2\alpha} + O(r^2) \right) \eta = 0, \quad (9)$$

$$\eta_{\pm} \sim r^{-5/2} e^{\pm 4\sqrt{6\alpha}/r}. \quad (10)$$

At infinity the two modes are r^2 and r^{-4} . A nonzero AF mode cannot select the decreasing inner exponential: multiplying (8) by η and integrating would give $\int (A\eta'^2 + V\eta^2) dr = 0$, since both boundary terms vanish. Thus its inner growing exponential is unavoidable. For a homogeneous outer amplitude δ , linear breakdown would occur at $r_{\text{layer}} \sim 4\sqrt{6\alpha}/\log(1/|\delta|)$, with logarithms of logarithms as corrections. This is conditional on the overlap and on δ ; the calculation does not determine $\delta(J)$.

For $u = 1 - N$, $r^2 = 2\alpha(1 - u^2)/u$, the exact rational equation is

$$\eta_{uu} + \frac{3u^3 + u^2 + 11u + 1}{u^4 - 1} \eta_u + \frac{2(u^4 + 10u^2 + 1)}{u^2(u-1)(u^2-1)^2} \eta = 0. \quad (11)$$

It has an irregular singular point at the physical endpoint $u = 1$.

The forced order- J^2 problem. The exact linear rotation in the critical round metric is

$$w(r) = \frac{4J}{\pi\mu} \frac{1 - N(r)}{r^2}, \quad w' = -\frac{16J}{\pi r^3 \sqrt{r^4 + 16\alpha^2}}, \quad w \sim \frac{2J}{\pi\alpha r^2} \quad (r \rightarrow 0).$$

Both derivative and endpoint normalizations were checked exactly. Evaluating this formula at $r^2 \sim g_H$ gives $w = O(J^{1/3})$ at fixed α ; it cannot alone normalize a finite inner angular velocity.

Writing $\eta = J^2 F(u)/(\pi^2 \alpha^3)$, the order- J^2 source on the right of the operator in (11) is

$$\frac{8u(3u^6 + 2u^5 - 7u^4 + 20u^3 + 5u^2 + 2u - 1)}{(u-1)^6(u+1)^5(u^2+1)}.$$

Its formal algebraic inner particular solution begins $\eta = -32J^2/(\pi^2 r^6) + \dots$, checked by substitution. Rotation therefore forces strong squashing at $r = O(J^{1/3})$ even if the homogeneous growing exponential is tuned away.

There is no rational particular solution in u . At $u = -1$ the homogeneous indicial roots are 1, -3 , whereas the source contains $3/[2(u+1)^5]$. A rational pole of order three cancels the needed order-five term, and other pole orders cannot match it. The required leading logarithm is $-3\log(u+1)/[8(u+1)^3]$, verified explicitly. This exclusion concerns only this forced outer equation; it is not a no-go theorem for the mass.

Inner action before exterior matching. For $\alpha = 1$, set $r = \epsilon R$, $g = \epsilon^2 G$, $h = \epsilon^3 H$, $b = \epsilon^3 B$, $f = \epsilon^2 F$, $w = W$. The leading action is

$$8\epsilon^3 \int dR \sqrt{FH/B} \left(\frac{B}{F} + \frac{B'H'}{H} + H(W')^2 \right).$$

Choosing $H = R$, its exact extremal BTZ family is

$$B = \frac{C^2(R - H_0)^2}{4R}, \quad F = (R - H_0)^2, \quad W = W_0 + \frac{CH_0}{2R}, \quad G(R) \text{ arbitrary.}$$

All leading Euler equations vanish. The conserved momentum gives $J = \pi\epsilon^3 H_0$, and the throat fixes $H_0 = 1/\sqrt{2}$ when $g_H = \epsilon^2$. The first G -dependent correction to the action is exactly $\epsilon^4 d[-2C(R^2 - H_0^2)/G]/dR$ on this family. Consequently these orders leave $\Omega_H = W_0 + C/2$ undetermined. Normalizing the exterior requires the nonlinear connection problem.

Nonlinear intermediate connection. Still at $\alpha = 1$, set $r = \epsilon R$, $(b, f, g, h) = \epsilon^2(B, F, G, H)$, $w = \epsilon W$, and $j = J/(\pi\epsilon^3)$. Writing $U = \sqrt{B}$ and $X = \sqrt{H}(4 - H/G)$, the exact leading reduced action is

$$L_0 = \frac{2UX}{\sqrt{F}} + 8\sqrt{F}U'X', \quad U = c\sqrt{X^2 - K}, \quad F = \frac{X^2 - K}{4(X')^2}.$$

All its Euler equations and the conserved Wronskian $8\sqrt{F}(U'X - UX') = 4cK$ were checked exactly. The round critical exterior has $U \sim R/2$, $X \sim 3R$, fixing $c = 1/6$. The zero leading value of K is appropriate when H shrinks at the core; its order- ϵ part is retained below. At fixed angular momentum,

$$W' = \left(\frac{2j}{3H} \right)' - \frac{4j}{3} \frac{G'}{G^2(4 - 3H/G)}, \quad (12)$$

$$W(R) = \frac{2j}{3H(R)} + \frac{4j}{3} \int_R^\infty \frac{G'(x) dx}{G(x)^2[4 - 3H(x)/G(x)]}. \quad (13)$$

If G is increasing and $0 \leq H/G \leq 1$, the residual integral stays bounded at the core. Matching $H_{\text{intermediate}} = \epsilon H_{\text{core}}$ then gives $W_0 \rightarrow 0$ and $B/H \rightarrow 4/9$, hence $C = 4/3$ and $\Omega_H \rightarrow 2/3$.

The next effective Routhian fixes the previously arbitrary squashing profile. Choose $G = R^2$, put $q = H/G$, $p = Rq'$, and define $A = 4 - q$, $T = 4 - 3q$, $D = 2qA + pT$. It is

$$\mathcal{L}_{\text{eff}} = R^3 \left[\frac{A(6q + p)}{12} + \frac{qA^2(3q + p)}{3D} - \frac{q^2A^3(2q + p)}{3D^2} \right] - \frac{16j^2D}{3R^3q^2T}. \quad (14)$$

Its Euler equation agrees with the independent forced exterior result: $q = 1 - 32z + 4096z^2 + \dots$, $z = j^2/R^6$. The regular core endpoint requires $z \rightarrow 1/2$, $p \rightarrow 4$. With q as independent variable its exact formal core recurrence starts

$$z(q) = \frac{1}{2} - \frac{3}{4}q - \frac{7}{32}q^2 - \frac{29}{24}q^3 + \dots$$

The full rational $z''(q)$ is recorded in `effective_core.json`.

Direct shooting was unstable from either endpoint and produced rejected profiles. A two-ended BVP succeeds after an exact recentering $z = z_{\text{polynomial}} + Y$ and Horner evaluation near each endpoint:

$[q_{\min}, q_{\max}]$	nodes	maximum RMS residual	$z > 0, z' < 0$
[0.005, 0.999]	823	1.983×10^{-8}	yes
[0.003, 0.9995]	862	1.994×10^{-8}	yes
[0.002, 0.9997]	906	1.990×10^{-8}	yes

On $q \in [0.02, 0.98]$, successive profiles differ by 8.249×10^{-12} and 7.266×10^{-12} . Positivity and monotonicity imply positive B, F, D and a bounded rotation integral in this reduced profile. Earlier unrecentered runs failed at tighter tolerances; their failures are retained in `effective_bvp.json`. These checks concern (14), not the full EGB boundary problem.

Matching the constant term of $B_{\text{BTZ}} = 4(H - H_0)^2/(9H)$ to the intermediate family gives $K = 32\epsilon H_0 + o(\epsilon)$. A round static mass perturbation has

$$\frac{N(\epsilon R; \mu = 2 + \epsilon^3 d)}{\epsilon^2} = \frac{R^2}{4} - \frac{\epsilon d}{4} + O(\epsilon^2), \quad d = \frac{32H_0}{9}.$$

Thus $\Delta M = 4J/3$ at $\alpha = 1$, independently consistent with $dM = 2\Omega_H dJ$. Restoring dimensions gives the conditional leading formula at the beginning and

$$m(s) = \frac{3s}{4} + \frac{4}{3\sqrt{s}} + o(s^{-1/2}).$$

The assumptions are: convergence to the round critical static exterior; an intermediate/core overlap with positive X, X' ; a uniformly bounded residual rotation integral; and no further layer changing the leading time normalization or mass mode. They are not established by the BVP.

A further conditional first correction. The converged intermediate profile gives the positive connection integral

$$I = \frac{1}{4} + 3 \int_0^1 \frac{[2z(q)]^{1/3}}{(4-3q)^2} dq = 0.537593821772 \dots$$

Three cutoff choices agree within 2.71×10^{-13} ; quadrature's reported error estimate is 3.22×10^{-12} . In a constant- G core, the first corrected lapse constraint gives $N_{\text{lapse}} = n_0(8/H - 6\epsilon)$ and $W' = -8n_0 j/H^2$ exactly through this order. *If additionally* the matching has $n_0 = 1/12 + O(\epsilon^2)$ and a regular first-order overlap, the throat series gives

$$\Omega_H \sqrt{\alpha} = \frac{2}{3} + \omega_1 \sqrt{g_H/\alpha} + o(\sqrt{g_H/\alpha}), \quad \omega_1 = \frac{4I}{3\sqrt{2}} - \frac{\sqrt{2}}{6} = 0.271146055470 \dots$$

It would produce a $J^{4/3}$ term in M and a $0.456526735220 \dots /s$ term in $m(s)$. This is a stronger conditional extension: the full first-correction EGB equations and uniform matching have not been verified.

Numerical probe, receipts, and omissions. A bounded isolated rerun at $a = 3.1175933011$ gave constraints 5.329×10^{-4} , 1.073×10^{-4} and 1.245×10^{-5} at $N = 64, 96, 144$, respectively. The $N = 96/144$ masses differ by 3.69×10^{-6} . The subsequent direct step to $a = 3.3$ failed: the $N = 96$ constraint was 40.14 and the $N = 144$ candidate had negative metric factors. These are rejected iterates, not new solutions. No production branch data were overwritten.

Commands use the existing Python 3.11.9 environment at `work/analytic_metric/.venv/Scripts/python.exe-B-u`; SymPy is 1.14.0. The scripts `check_endpoint.py`, `static_squashing.py`, `forced_squashing.py`, `inner_btz.py`, `indicial_crossing.py`, `matching_constants.py`, `effective_bvp_stable.py`, and `core_frame_shift.py` write matching JSON receipts with actual output. Selected actual lines:

```
PASS inherited indicial roots: 63 points, max absolute difference 1.981e-13
PASS all five linearized EOM: b,f,w vanish; g,h give exactly the NVE
PASS leading forced IR particular eta = -32J^2/(pi^2 r^6)
PASS first G-dependent correction is d[-2C(R^2-H0^2)/G]/dR on BTZ
PASS leading intermediate general family U=c*sqrt(X^2-K), F=(X^2-K)/(4Xp^2)
```

Not checked: uniform nonlinear overlap, the amplitude $\delta(J)$, global convergence to the proposed endpoint, or continuation to arbitrary a . The displayed endpoint expansion is conditional. No finite-coupling mass formula was obtained. No figure was reproduced. No commit or push was made.

References

- [1] Y. Brihaye, B. Kleihaus, J. Kunz and E. Radu, arXiv:1010.0860, eqs. (2.11)–(2.12). Local PDF: `papers/1010.0860-brihaye-kleihaus-kunz-radu-equal-spin-egb-d5.pdf`.
- [2] B. Kleihaus, J. Kunz and E. Radu, arXiv:2303.12471, secs. 2–3. Local PDF: `papers/2303.12471-kleihaus-kunz-radu-phases-rotating-d5-egb.pdf`.